Edward Ajayi

+250795008889 | eaajayi@andrew.cmu.edu | GitHub | Google Scholar
https://edwardajayi.github.io/

Education

Carnegie Mellon University
Master of Science in Engineering Artificial Intelligence — *Aug. 2024 – May. 2026*

- **Relevant Courses:** Natural Language Processing, Deep Learning, Question Answering with LLMs
- **Specializations:** Data Science, Machine Learning, NLP
- **GPA:** 3.73/4.0

Ekiti State University
Bachelor of Science in Computer Science — *Oct. 2014 – Nov. 2019*

- **Specializations:** Computer Science
- **GPA:** 4.17/5.0

Research Interests

Language Modeling, Representation Learning, Multimodal Reasoning, Affective Computing, Socially Aware AI

Technical Skills

Languages: Python, SQL, Bash
Generative AI & LLMs: LangGraph, OpenAI Agents SDK, RAG, LLM Fine-Tuning, Prompt Engineering, DPO
Deep Learning & ML: PyTorch, Hugging Face Transformers, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn
Natural Language Processing: spaCy, NLTK, Sentiment Analysis, Text Processing
Data Engineering & Ops: Docker, AWS, FastAPI, Neo4j, BeautifulSoup, Jupyter

Selected Publications

1. **E. Ajayi**, M. Kachweka, M. Deku, and E. Aiken. *A Machine Learning Approach for Detection of Mental Health Conditions and Cyberbullying from Social Media. Proceedings of Machine Learning Research (PMLR), AAAI-26 AI in Medicine and Healthcare Bridge*. **Best Paper Award**. 2026. [PMLR]
2. **E. Ajayi** and P. Mitra. *HumorGen: Cognitive Synergy for LLM Humor Generation via Persona-Based Distillation*. Submitted to *EMNLP*. 2026. [arXiv Preprint]
3. **E. Ajayi** and P. Mitra. *HumorRank: An Automated Leaderboard for Evaluating Humor Generation in Large Language Models*. Submitted to *COLM*. 2026. [Manuscript]
4. **E. Ajayi** and P. Mitra. *Automatic Humor Detection: From Theoretical Foundations to Large Language Models*. Under Review. 2025. [Manuscript]
5. **E. Ajayi**, B. Tadele, E. Umwari, M. Deku, P. Singadi, C. Edeh, and J. Udahemuka. *Language Diversity: Evaluating AI Performance on African Languages*. 2025. [arXiv Preprint]
6. **E. Ajayi**. *AfriEconQA: A Benchmark Dataset for African Economic Analysis Based on World Bank Reports*. Under Review. 2026. [arXiv Preprint]

Experience

Research Associate
Spatial and Language Technologies Lab, Carnegie Mellon University Africa — *Jun. 2026 – Present*

- Building efficient multimodal humor generation models using LLMs and vision-language models (VLMs).
- Developing evaluation benchmarks for Large Language Models on subjective generation tasks and alignment.
- Conducting research on computational humor, multilingual NLP, and reasoning abilities of foundation models.

Predoctoral Research Fellow
Human-Algorithm Interaction Lab, Oxford Saïd Business School, University of Oxford, UK — *Feb. 2026 – May. 2026*

- Conduct research on Human-AI collaboration, decision-making, and AI adoption in service systems at the Human-Algorithm Interaction Lab.

Research Assistant
Spatial and Language Technologies Lab, CMU — *Aug. 2025 – May. 2026*

- Conducting research on NLP for automatic humour detection and generation, dataset preparation, model training, evaluation, and benchmarking.
- Contributing to multilingual and low-resource language technologies focused on culturally grounded language models.

Research Intern - Business Automation
Carnegie Mellon University — *Jun. 2025 – Aug. 2025*

- Built an agentic AI workflow framework using LangGraph and FastAPI to automate complex business workflows.

- Researching and implementing AI safety protocols and robust LLM reasoning for ethical alignment and deployment.
- Analyzing agent performance to establish benchmarks for human-in-the-loop oversight and develop explainability metrics to support transparent corporate accountability.

Client Service Officer
Stanbic IBTC Bank *Dec. 2021 – Jul. 2024*
- Leveraged data-driven insights to cross-sell over N400M in assets, enhancing client trust and digital adoption by 25%.
- Managed client relationships, identifying service gaps to improve operational efficiency.

Selected Projects

HumorRank — LLM Evaluation & Ranking Framework | *Python, Evaluation* Leaderboard [Paper]
- Developed a scalable evaluation framework for ranking outputs from multiple LLMs on subjective generation tasks.
- Implemented pairwise comparison with probabilistic ranking using the Bradley-Terry model.
- Designed a candidate generation → evaluation → leaderboard workflow for scalable benchmarking.

HumorGen-7B — LLM-based Humor Generation Model | *PyTorch, Transformers* HuggingFace
- Developed HumorGen-7B, a fine-tuned 7B language model for controlled humor generation.
- Implemented end-to-end alignment pipeline using supervised fine-tuning and Direct Preference Optimization (DPO).
- Implemented Cognitive Synergy framework to induce diverse reasoning paths for humor generation.

MyTorch — Minimal Deep Learning Framework Built from Scratch | *Python*
- Implemented core tensor operations, automatic differentiation, activation layers, and SGD without deep learning libraries.
- Designed a PyTorch-style API enabling end-to-end model training while illustrating neural network internals.
- Built modular components for educational experimentation with deep learning systems.

Mental Health & Cyberbullying Detection System (AAAI Oral) | *Transformers* GitHub [arXiv]
- Built an end-to-end multi-label classification system for detecting mental health signals and cyberbullying from social media data.
- Combined classical ML methods with transformer embeddings for robust prediction on noisy datasets.
- Produced results leading to an Oral paper acceptance at AAAI-26 AIMedHealth.

AI Workflow Automation Agent | *Python, LangChain, FastAPI*
- Built an autonomous agent to process submissions, route tasks, and trigger approval workflows.
- Integrated API orchestration, notifications, and real-time validation pipelines.

Training & Conferences

Oral & Poster Presentation, AAAI Conference on Artificial Intelligence Singapore
AAAI *Jan. 2026*
- Presented my paper on ML for mental health and cyberbullying detection from social media, and engaged with academics and researchers at the conference.

Participant, Deep Learning Indaba 2025 Rwanda
Deep Learning Indaba *Aug. 2025*
- Attended workshops on deep learning, AI for social good, and research collaboration across African universities.

Participant, Digital Humanities Summer School Rwanda
DIGIHUM/CMU-Africa *Jul. 2025*
- Engaged in computational methods for humanities research, including NLP applications.

Volunteering/Leadership

President, CMU-Africa Research Club
Carnegie Mellon University *Jan. 2025 – Dec. 2025*
- Led 12 major events in 2025 supporting student research.
- Launched website with research guides, conference updates, and collaboration hub.
- Organized first Student Research Showcase with 20+ posters and workshops.
- Hosted three-part PhD Application Series, faculty sessions, and alumni panels supporting 40+ students.
- Collaborated with ACM SIGCOMM and Upanzi Network for remote conference participation.
- Built an active research ecosystem that provided critical peer support, contributing to over 20 student acceptances to top-tier venues including NeurIPS and AAAI in 2025.

Reviewer, AAAI AIMedHealth Bridge 2026
AI for Medicine and Healthcare *Nov. 2025 – Nov 2025*
- Reviewed submissions in applied machine learning and AI for healthcare under the AAAI AIMedHealth Bridge 2026 program.

IEEE Student Member, CMU-Africa
Institute of Electrical and Electronics Engineers *Feb. 2025 – Present*
- Collaborating with IEEE members to advance research initiatives among club members.

Certifications

- Microsoft Certified Azure Data Scientist Associate - View
- DataCamp Python Data Associate - View

Honors and Awards

- Mastercard Foundation Scholarship, Carnegie Mellon University Africa
- Federal Scholarship Board Award for Undergraduates, Ekiti State University